

Trevon Woods
Sitaram Ayyagari
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L02 Reflective Journal

This journal is a reflection of my experience throughout this lab. I chose to do Option B for this lab as I am a very hands on person. I will try my best to convey any challenges I may have faced, what I learned, and my understanding of how Tensorflow Lite applies to real-world A.I. deployments.

This was a very informative lab, I learned a few things that I didn't know before and got to add a new deployment implementation for A.I. models. First things first, I liked doing research and learning about Tensorflow Lite. I had heard about it before but never really looked that deep into it. I understand now just how important this module is, mainly due to the need to deploy A.I. models on edge devices is such a prevalent area of study these days. I especially enjoyed the inference part of the lab because it put into perspective what needs to go on behind the scene. We have to first load the model via the `model_path`, then we have to allocate memory for all of the input and output tensors as well as activations between layers. This opened my eyes to the fact that on edge devices there is only so much room. A developer must take into consideration how much memory the model needs for these things. Next, we needed to ensure the d-type of our inputs match the models expected d-type along with reshaping the input to ensure it also aligns with what our model expects. Then we had to load the input into the model by directly indexing into its memory buffer before invoking the model to obtain its prediction. I truly enjoyed this lab because now I know how to use this Tensorflow Lite module to convert my regular Tensorflow models into models I can use on an edge device.

The only challenge I had throughout this lab was understanding what was going on when using the Tensorflow Lite model for inference. For example the `.set_tensor()` function threw me off a little. At first I was under the impression this was similar to randomly initializing a model with weights. After some research I was able to understand that it was simply just a way to load the input into the model for its use in the model's inference. Also the `.allocate_tensors()` function was also something I had to look into because of the same thinking I had with the `.set_tensor()` function. I completely understand what they do now and have a strong understanding of how Tensorflow Lite works and look forward to using it in projects in the future.